

Luminary

Cross-University Research Discovery Platform Product Management Case Study

Uttkarsh Sharma | Team Lead | Protothon 8.0, APMC @ Northeastern | March 2026

1st of 50 teams · 2-week build · 6 simulated university nodes · privacy-by-architecture design

This document covers the problem discovery, product decisions, prioritization, metrics framework, and learnings from building Luminary — a federated research discovery platform that makes institutional research visible across university boundaries without moving raw data.

Executive Summary

Research universities collectively produce billions in research output each year, yet the infrastructure for researchers to find each other's work, avoid duplication, and form collaborations is largely absent. Luminary is a cross-university research discovery platform that makes ongoing institutional research visible and collaborable without moving raw data across institutional lines.

AREA	SUMMARY
Problem	PhD researchers duplicate months of work, miss relevant collaborators, and wait weeks for cross-institutional data access. Three of three interviewees reported waits of 4–12 weeks.
Solution	A federated platform combining RAG semantic search with QAOA-based complementarity matching. Raw data stays on university servers; only model weights and embeddings move. Compliance posture is designed into the architecture rather than promised in policy.
My role	Team Lead on a 4-person team (including me) at Protothon 8.0. Owned product strategy, user research, RICE prioritization, metrics design, go-to-market, and the QAOA implementation.
Result	1st of 50 teams. 3 user interviews; 5 failure modes surfaced, 3 confirmed unanimously and 2 by majority. Functional prototype built in 2 weeks.

Market context

FIGURE	BASIS
\$28B / yr	US spend on preclinical research that cannot be reproduced (Freedman, Cockburn & Simcoe, <i>PLOS Biology</i> , 2015). Duplicated effort is an adjacent, less-measured cost — this figure frames the scale of waste, it does not measure duplication directly.
7.2 yrs	Median time from graduate school entry to a US research doctorate, all fields, 2022 — down from 8.6 years in 2002 (NSF Survey of Earned Doctorates). Months lost to duplicated work are months added to an already long degree.
8 institutions	R1-class research universities within roughly five miles in Boston, with no shared discovery infrastructure connecting their researchers.

Problem Statement

The problem is not that researchers are unmotivated. It is that discovery infrastructure has not scaled with the volume of research being produced. Boston has eight world-class research institutions within five miles of each other and no shared infrastructure connecting their PhD students.

Five failure modes

Across 3 user interviews (7 open-ended questions each), the same failure modes recurred without leading questions. Three were reported by all three participants; two by two of three.

FAILURE MODE	REPORTED EXAMPLE	
Duplication of effort is the norm	3 / 3	Arjun (NEU) spent 4 months on a federated learning pipeline. A lab two buildings away had finished the same work 6 months earlier.
Discovery is accidental	3 / 3	Wei (MIT) met his ideal collaborator at a conference, two years after they could have connected.
Data access takes 4–12 weeks	3 / 3	Priya (Tufts) emailed 12 professors. Eight weeks later she was still waiting; her thesis deadline moved.
Dead ends repeat across cohorts	2 / 3	No institutional memory. Each cohort rediscovers the same failed approaches.
Collaboration arrives too late	2 / 3	Methodologies built in isolation that would have been stronger with earlier input.

User voices

Arjun — PhD Candidate, Northeastern

“The hardest part is figuring out what work already exists. A lot of time is spent searching scattered sources.”

Priya — PhD Researcher, Tufts

“Projects die on the vine because the administrative friction of moving data across departments is higher than the value of the research.”

Wei — PhD Student, MIT

“If I had known they were working on that algorithm last year, my methodology would have been far stronger.”

Sample size is the main limitation of this research. Three interviews are enough to surface a pattern and shape a two-week build; they are not enough to size a market or claim external validity. The next step would be 15–20 interviews across research-intensive and teaching-intensive institutions.

Key Product Decisions

A two-week build forces clarity on what the product actually is. Four decisions defined Luminary's direction, including one explicit scope cut.

01 Federate at the institution level, not the individual level

Context	The obvious design was peer-to-peer connections — a LinkedIn for research. Wrong unit of analysis.
---------	--

Problem	Universities own the IP. Individuals cannot grant access to institutional datasets.
---------	---

Decision	Made universities the federation unit. Each runs a node; Luminary connects nodes.
----------	---

Outcome	Unlocked the enterprise sales motion, the compliance story, and network effects at scale.
---------	---

02 Design compliance into the architecture, not the policy

Context	Competitors trade privacy for comprehensiveness, or comprehensiveness for privacy.
---------	--

Problem	University IT has seen privacy-first platforms fail on a policy promise.
---------	--

Decision	Raw research data and student records never leave the institution's node. FERPA applies where student records are involved and HIPAA where clinical datasets are; both are addressed by keeping the regulated data in place.
----------	--

Outcome	Shifts procurement from 'trust our policy' to 'the data does not move.' Residual model-layer risk is addressed in Section 09.
---------	---

03 Optimize for complementarity, not similarity

Context	RAG surfaces similar researchers. Similarity is not what drives strong collaboration.
---------	---

Problem	A researcher on the same problem is a competitor. Complementarity is the useful signal.
---------	---

Decision	Two layers: SciBERT embeddings with cosine similarity for discovery; a QUBO formulation solved with QAOA for complementarity scoring.
----------	---

Outcome	Differentiated position — existing tools rank by similarity alone.
---------	--

04 Simulate the quantum components (deliberate scope call)

Context	Running on real quantum hardware requires NISQ-era stability not available at prototype scale.
---------	--

Problem	Overpromising hardware readiness creates technical debt and credibility risk in a 2-week sprint.
---------	--

Decision	Classical simulation of QAOA and the encoding scheme. Architecture is quantum-ready; flows are correct.
----------	---

Outcome	Working prototype in 2 weeks, structured for hardware migration when the hardware matures.
---------	--

The Solution

Luminary is a three-layer federated system. The architecture resolves the central tension in research collaboration: comprehensiveness and data sovereignty cannot coexist in a centralized database. Luminary brings the intelligence to the data rather than the data to the intelligence.

RAG finds the researchers most *similar* to a query. QAOA finds the researchers most *complementary* to the work. Those are different objectives, and complementarity is the one that predicts a productive collaboration.

Four capabilities

CAPABILITY	TECHNOLOGY	WHAT IT DOES
Semantic discovery	SciBERT embeddings + cosine similarity	Retrieval-augmented search across connected university nodes. Surfaces in-progress research, not only published papers.
Local encoding	Walsh–Hadamard + phase encoding + CNOT (classically simulated)	Data is encoded locally before transmission; the institution's key is required to decode.
Complementarity matching	QUBO formulation solved with QAOA (classically simulated)	Treats collaborator pairing as a combinatorial optimization over four variables rather than a similarity ranking.
Federated learning	Gradient-only training across nodes	Only model weights travel the network. Raw research data and IRB-protected datasets stay on the node.

What this changes

Without Luminary, a PhD student spends weeks searching scattered sources, emailing professors across universities, and waiting on approvals — only to discover months later that the work already existed. With Luminary, a query returns ranked researchers across connected institutions in minutes, with complementarity scores showing not just who works on similar problems but who would be most useful to work with. Duplication is caught before it becomes a sunk cost.

Status of the prototype. Six university nodes were simulated on sample data. The quantum components are classically simulated. What was demonstrated is that the flows are correct end to end, not that the system has run in production.

Prioritization

With a two-week window and a team of four, every feature had to earn its place. I scored candidates with RICE — $\text{Reach} \times \text{Impact} \times \text{Confidence} \div \text{Effort}$ — using a 1–10 scale for all four factors rather than the conventional percentage confidence and 0.25–3 impact scales. The absolute numbers are therefore only meaningful relative to each other within this table.

FEATURE / DECISION	R	I	C	E	RICE	CALL
RAG semantic search (SciBERT + cosine)	9	9	9	5	146	Built
Federated architecture (institution nodes)	9	10	9	8	101	Built
QAOA complementarity matching	8	10	8	7	91	Built
Local encoding (classically simulated)	7	9	8	6	84	Built
P2P direct messaging between researchers	6	5	7	4	53	Cut
GDPR compliance (EU partners)	4	7	6	8	21	Phase 3
Real quantum hardware integration	5	8	3	10	12	Cut

The two cuts

P2P messaging (53). Cut because it solves a communication problem and the validated pain point is discovery. The counter-argument is real and worth stating: participants described eight-week email waits, which sounds like a messaging problem. It is not. They were waiting on *data access approvals*, not on a channel to send a message through — every one of them already had email. Adding a second inbox does not shorten an approval queue. What shortens it is knowing, before you ask, who actually holds relevant work and whether it is worth the request. Messaging is the right Phase 2 feature once discovery works.

Real quantum hardware (12). Cut on confidence and effort, not importance. NISQ-era stability is years out; the prototype's value proposition does not depend on it, and building on unstable infrastructure produces a demo that impresses once and fails under scrutiny.

Metrics Framework

For a B2B platform with network effects, the North Star has to capture individual researcher value and institutional value at once, and grow as the network expands.

North Star: collaboration requests initiated per month

Directly measurable, and a proxy for the value Luminary creates: a request is a researcher acting on a match rather than merely viewing one. Each additional node expands the match space for every existing node, so the metric compounds with the network. This is a production metric — at prototype scale the observable proxy was searches run and matches surfaced against sample data.

AARRR funnel with Phase 1 targets

STAGE	METRIC	TARGET	RATIONALE
Acquisition	Universities onboarded via Research VP outreach and conference presence	3 universities	Boston cluster strategy. Launching three at once addresses cold start; each node lifts the others.
Activation	Researchers completing a search within 7 days of onboarding	60%	The first search is the activation event — one search means the core value has been experienced.
Retention	Researchers returning to search again within 30 days	40%	Research runs over months. Thirty-day return suggests the tool is embedded in a real workflow.
Revenue	SaaS licensing per university (Tier 2)	\$150K ARR, Year 1	One paying institution at Tier 2. The \$1.5M figure is a Year 2+ network target, not a Year 1 target.
Referral	Universities referring peer institutions through consortia	2 per university	Each node has a structural incentive to grow the network — more nodes means better matches.

Targets are hypotheses, not forecasts. The 60% and 40% figures are benchmarks borrowed from B2B SaaS activation and retention norms, set as something to test in the Phase 1 pilot rather than something already observed.

Competitive Analysis

Existing solutions fail along the same axis: they trade data sovereignty for comprehensiveness, or comprehensiveness for sovereignty. None addresses discovery of *in-progress* research across institutional boundaries. That gap is Luminary's position.

SOLUTION	IN-PROGRESS WORK	CROSS-INSTITUTION	SEMANTIC SEARCH	DATA STAYS ON SITE	COMPLEMENTARITY SCORE
Google Scholar	No	Published only	Keyword	n/a — public data	No
ResearchGate	No	Yes	Limited	Centralized profile data	No
Semantic Scholar	No	Yes	Yes	n/a — public data	No
Elsevier Pure / SciVal	Partial	Within-institution silo	Keyword-led	Vendor-hosted	No
Centralized shared DB	Yes	Yes	Yes	No	No
Luminary	Yes	Yes	Yes (+ federated)	Yes, by architecture	Yes (QAOA)

Why the architecture is the moat

A centralized database requires every institution to surrender data sovereignty. That single requirement has blocked cross-university research infrastructure for decades. Luminary never asks for it, because no raw data moves — what moves is model weights, embeddings, and scores. Each university keeps control of its research records and protected datasets.

The defensibility claim should be stated carefully. A federated architecture is not technically impossible for an incumbent to build; the barrier is that a centralized platform's data model, contracts, and revenue structure all assume the data is theirs to hold. Rebuilding around federation means renegotiating the existing book of business. The moat is commercial inertia plus signed node agreements, not a technical secret — and the honest read is that it buys years, not permanence.

Business Model

Luminary is B2B SaaS. Economic buyers are Research VPs and Provosts at research-intensive universities; department heads are the internal champions; PhD researchers are the end users. Pricing is benchmarked against Elsevier SciVal and Clarivate InCites, which sit in the low-to-mid six figures annually.

Market sizing

	SIZE	BASIS
TAM	~\$2B	Research intelligence and analytics software, estimated. This is a directional figure built from incumbent pricing and institution counts, not a purchased analyst estimate.
SAM	~\$50M	Roughly 500 R1-equivalent institutions worldwide at ~\$100K average. The US alone has ~146 Carnegie R1 universities, a ~\$14.6M ceiling — the market only becomes interesting internationally.
SOM	~\$1.5M	Ten Boston-area institutions at \$100–200K per year, reachable in Year 2+ if the cluster strategy works.

Pricing tiers

TIER	PRICE	SCOPE	BUYER
Tier 1 — Department pilot	\$25K / yr	One department, three systems. Free pilot option to remove the procurement barrier.	Department Head
Tier 2 — University licence	\$100–150K / yr	Full institution, all departments, cross-university discovery active.	Research VP / Provost
Tier 3 — Network licence	\$200K+ / yr	Multi-university node with full network effect; consortium pricing.	University Consortium

Unit economics (Tier 2). Revenue \$100K/yr against roughly \$5K/yr infrastructure gives a ~95% gross margin. Customer acquisition cost is not modelled here — enterprise university sales cycles run 6–18 months and carry real cost, so CAC payback is an open question rather than a strength.

Go-to-market phases

PHASE	TIMELINE	MILESTONE	REVENUE
Phase 1 — Department	Months 1–6	Single department pilot, ~100 researchers. Gate to proceed: 3 cross-department collaborations initiated.	Free pilot
Phase 2 — Full university	Months 7–12	All colleges live; dataset visibility and grant landscape features shipped.	\$150K ARR
Phase 3 — Network	Year 2+	Additional Boston institutions onboarded as nodes; network effect active.	~\$1.5M ARR

Risk Analysis

Five risks were identified and explicitly mitigated. Raising them with specific mitigations builds more credibility than avoiding them; each one shaped an architectural or go-to-market decision.

RISK	SEVERITY	MITIGATION
University adoption. Procurement is slow; central IT approval can run 6–18 months.	High	Start with department heads rather than central IT. A free pilot removes the budget barrier and faculty champions carry it internally.
Cold start. One or two universities gives insufficient match density to be useful.	High	Pre-load public thesis and publication data. Onboard three institutions simultaneously via the Boston cluster.
Data privacy. A single breach would end institutional trust permanently.	High	Regulated data stays on the node. Third-party security audit before launch; IRB advisory input from day one.
Model-layer attack. Federated learning protects raw data but not gradients — inversion attacks can reconstruct training data without touching the source.	High	Differential privacy on gradient updates is a prerequisite for production. Not implemented at prototype stage, which is why the compliance claim in this document is scoped to data movement rather than absolute security.
Quantum dependency. QAOA and encoding are classically simulated.	Medium	Simulation is architecturally correct at demo scale, and the product's value does not depend on quantum hardware.
Incumbent response. Elsevier or Microsoft could attempt to replicate the model.	Medium	Signed node agreements and the incumbents' own centralized revenue structure slow a response, but do not prevent one.

Risk posture

Three of the four high-severity risks are trust problems rather than technology problems, and the same architectural decision addresses all three: keep the regulated data where it already sits. That changes the procurement conversation from 'trust our policy' to 'the data does not move.' The fourth — gradient inversion — is a genuine technical gap, and the honest position is that Luminary is not production-ready for regulated data until differential privacy is in place.

Results & Reflections

Outcomes

1st of 50 teams	Protothon 8.0, APMC @ Northeastern, March 2026.
3 interviews	Five failure modes surfaced; three confirmed by all three participants, two by two of three.
$R^2 = 0.85$	Proportion of variance in the match-score target explained by the SciBERT similarity model on a held-out split of the simulated dataset. This is a pipeline sanity check — see the note below on what it does and does not establish.
2-week build	Federated architecture, RAG search, QAOA matching, and local encoding, end to end.
~95% gross margin	Modelled, not observed: ~\$5K/yr infrastructure against \$100–150K annual licence.

What $R^2 = 0.85$ does and does not establish. R^2 is the share of variance in a target variable that a model accounts for — 0.85 means 85% of the variation in the match-score target is explained by the model's inputs. On real labelled outcomes that would be a strong result. Here the labels came from the same simulated dataset the nodes were built on, so the figure measures internal consistency of the pipeline, not the quality of real-world matching. It establishes that embeddings, retrieval, and scoring are wired together correctly and that the model is not fitting noise. It does not establish that Luminary predicts productive collaborations. Doing that requires ground truth the prototype never had: human relevance ratings from researchers, or downstream co-authorship and grant outcomes tracked over years. That is the validation study, and it is the honest next step.

What I would do differently

Research depth before conviction. Three interviews shaped a two-week build, and that was the right trade under the constraint — but I wrote a market thesis on top of a sample of three. With more time I would run 15–20 interviews spanning R1 and non-R1 institutions before making sizing claims, and I would test whether the pain holds outside a dense research cluster like Boston.

Security at the model layer. Federated learning protects raw data but does not address gradient inversion. Differential privacy on gradient updates is a hard prerequisite before any production deployment involving protected data, and I should have scoped the compliance language around that limitation from the start rather than after the fact.

Cold-start sequencing. The Boston cluster strategy is directionally right, but I did not validate the sequencing: which department at which institution to sign first, what the free pilot has to demonstrate to convert, and who inside a university actually owns the budget line.

Key Takeaways

Luminary works as a case study because the research collaboration problem turned out not to be a technology problem. Every existing solution centralizes data in order to produce intelligence. The more useful move is to decentralize the data and centralize the intelligence — and that inversion, not any single feature, is what makes the product defensible.

The unit of analysis matters more than the feature set

Moving the federation unit from researcher-to-researcher to university-to-university unlocked the enterprise model, the compliance story, and the network effect at the same time. A better feature on the wrong unit of analysis would still have failed. Scope the right entity before designing the right features.

Technical fluency is knowing why a tool fits, not that it exists

QAOA earns its place only because complementarity matching is a combinatorial optimization that classical solvers handle poorly at scale. Naming an algorithm is not fluency; knowing the point at which the conventional approach breaks down is. The same standard cuts the other way — it is why the hardware integration was cut and why this document labels the simulated components as simulated.

Scope discipline is a product skill

Cutting real quantum hardware was right not because it was unimportant but because the value proposition did not depend on it. Knowing the boundary between an architecturally correct simulation and a production implementation is the difference between shipping something that works and shipping a demo that impresses once.

People problems and product problems are not the same problem

The strongest signal in the interviews — eight weeks waiting on an email — pointed at a feature that would not have helped. Approval friction is an organizational problem; the product problem underneath it was that researchers could not tell which requests were worth making. Cutting messaging was the prioritization decision I would defend hardest.

Uttkarsh Sharma | sharma.utt@northeastern.edu | [linkedin.com/in/01uttkarsh](https://www.linkedin.com/in/01uttkarsh) | portfolio-uttkarsh.lovable.app

MS Engineering Management | Northeastern University